

Act I · Develop

Principal Investigator

140 research points

Baby Mateo case · neural-crest spacing in the developing frontal bone

Student: **Jordan Rivera (sample student)**

Date: Example run, Act I

01 · QUESTION (ASK)

Do neural crest cells space themselves non-randomly as they build the frontal bone?

02 · HYPOTHESIS & PREDICTION

A graded FN1 “road” directs neural crest migration, producing ordered nuclear spacing.

PREDICTION FN1-rich tissue shows larger, more regular spacing than FN1-blocked control.

03 · TOOLS & BUDGET

Tools selected	Spatial layout
Evidence ceiling	Spatial layout
Team hired	none
Grant spent	\$0 of \$50,000

04 · DESIGN

Control group	on
Replicates	3 embryos
Honesty label	2D-projected (labeled honestly)

05 · MEASUREMENTS (RUN)

FN1-rich tissue · mean spacing (rel.)	0.0843
FN1-blocked (control) · mean spacing (rel.)	0.0450
Replicates (real n)	3
Honest per-replicate test	p < 0.001 · significant

06 · ANALYSIS

Only the 3 replicates count as the real sample (n=3), and the honest per-replicate test already finds a clear difference (p < 0.001). Pooling treats all 599 cells across those replicates as if each were its own experiment, inflating the sample far past 3. That inflation is what makes a pooled result look more certain than the evidence supports, which is pseudoreplication, so the per-replicate test is the one to

trust. Spacing here is a relative nearest-neighbor index in normalized image units; a larger number means cells sit farther apart.

07 · CONCLUSION · CLAIM · EVIDENCE · REASONING · LIMITATION

SUPPORTED

CLAIM Nuclei are non-randomly spaced in FN1-rich tissue compared to control.

EVIDENCE Nearest-neighbor spacing (relative, normalized units; larger = farther apart), FN1-rich tissue vs FN1-blocked (control), measured per replicate (n=3). Honest test $p < 0.001$; mean 0.0843 vs 0.0450.

REASONING The evidence reaches the Spatial layout rung, and this claim sits at or below that rung. A stronger claim would need a tool this lab does not have, so the run stops here.

LIMITATION Spatial organization is not the same thing as tension. Even with more money, you can't make a tension claim here: the machine isn't in this lab, and your embryo is alive.

08 · WHAT I CORRECTED (THE 6 RS · 2 STEPS REVISED)

Ask

REVISED I started with "why do babies get clefts," but that is a why I cannot measure. I changed it to whether the crest cells space themselves non-randomly, which I can actually count.

NEXT Write the testable version of my question first, before I get attached to the big one.

Conclude

REVISED My first instinct was to say FN1 pulls the cells into place. I changed it to "the cells are non-randomly spaced," because I measured spacing, not force.

NEXT Match the claim to the tool I actually used, and name out loud the thing I did not measure.

BADGES EARNED

Did the Reading · 15

Sharp Question · 10

Prediction Made · 10

Tooled Up · 10

Fair Design · 15

Verified Measurement · 15

Honest Ceiling · 25

Self-Corrector · 20

Closed the Loop · 20